PVsyst - Simulation report

Grid-Connected System

Project: BIPV_PV_Projet

Variant: Nouvelle variante de simulation

Building system

System power: 857 kWp

Suissi - Morocco

Author

PVsyst V8.1.2
VC0, Simulation date:
04/06/26 19:04
with V8.1.2

Project summary

Geographical Site	Situation		Project settings	
Suissi	Latitude	33.9805 °(N)	Albedo	0.20
Morocco	Longitude	-6.8569 °(W)		
	Altitude	54 m		
	Time zone	UTC		

Weather data
Suissi
Meteonorm 9.0 dll, Sat=100 % - Synthétique

System summary

Grid-Connected System — **Building system**

Orientation #1		Orientation #2		Orientation #3	
Fixed plane		**Fixed plane**		**Fixed plane**	
Tilt/Azimuth	30 / -51 °	Tilt/Azimuth	90 / -51 °	Tilt/Azimuth	90 / 39 °

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	4463 units	Nb. of units	32.1 units
Pnom total	857 kWp	Total power	751 kWac
		Pnom ratio	1.14

Results summary

Simulation step	Hourly				
Produced Energy	910054 kWh/year	Specific production	1061 kWh/kWp/year	Perf. Ratio PR	67.02 %

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General parameters

Grid-Connected System | **Building system**

Simulation step
Hourly

Orientation #1

Fixed plane		**Sheds configuration**		**Sizes**	
Tilt/Azimuth	30 / -51 °	Nb. of sheds	10 units	Sheds spacing	0.00 m
		Set of tables		Sensitive width	0.00 m
		Shading limit angle			
		Limit profile angle	°		

Orientation #2

Fixed plane		**Sheds configuration**		**Sizes**	
Tilt/Azimuth	90 / -51 °	Nb. of sheds	4 units	Sheds spacing	0.00 m
		Set of tables		Sensitive width	0.00 m
		Shading limit angle			
		Limit profile angle	°		

Orientation #3		**Models used**		**Horizon**
Fixed plane		Transposition	Perez	Free Horizon
Tilt/Azimuth	90 / 39 °	Diffuse	Perez, Meteonorm	
		Circumsolar	separate	

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

PV Array Characteristics

Array #1 - Champ PV tiotures
Orientation 1
Tilt/Azimuth 30/-51 °

PV module		**Inverter**	
Manufacturer	Generic	Manufacturer	Generic
Model	Mono 440 Wp Twin 144 half-cells	Model	350 kWac inverter, 12 MPPT
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	440 Wp	Unit Nom. Power	350 kWac
Number of PV modules	936 units	Number of inverters	13 * MPPT 8% 1.1 unit
Nominal (STC)	412 kWp	Total power	379 kWac
Modules	39 string x 24 In series	Operating voltage	480-1500 V
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.09
Pmpp	374 kWp	No power sharing between MPPTs	
U mpp	899 V		
I mpp	416 A		

Array #2 - Champ BIPV -51
Orientation 2
Tilt/Azimuth 90/-51 °

PV Array Characteristics

PV module		**Inverter**	
Manufacturer	Generic	Manufacturer	Generic
Model	a-Si:H, tripple junction	Model	12 kWac inverter with 2 MPPT
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	136 Wp	Unit Nom. Power	12.0 kWac
Number of PV modules	1903 units	Number of inverters	35 * MPPT 50% 17.5 units
Nominal (STC)	259 kWp	Total power	210 kWac
Modules	173 string x 11 In series	Operating voltage	350-600 V
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.23
Pmpp	252 kWp	No power sharing between MPPTs	
U mpp	357 V		
I mpp	705 A		

Array #3 - Champ BIPV 39

Orientation	3
Tilt/Azimuth	90/39 °

PV module		**Inverter**	
Manufacturer	Generic	Manufacturer	Generic
Model	Micro-crys/Amorphous 115Wc	Model	12 kWac inverter with 2 MPPT
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	115 Wp	Unit Nom. Power	12.0 kWac
Number of PV modules	1624 units	Number of inverters	27 * MPPT 50% 13.5 units
Nominal (STC)	187 kWp	Total power	162 kWac
Modules	203 string x 8 In series	Operating voltage	350-600 V
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.15
Pmpp	175 kWp	No power sharing between MPPTs	
U mpp	337 V		
I mpp	519 A		

Total PV power		**Total inverter power**	
Nominal (STC)	857 kWp	Total power	751 kWac
Total	4463 modules	Nb. of inverters	33 units
Module area	8505 m²		0.9 unused
Cell area	7532 m²	Pnom ratio	1.14
		No power sharing	

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.10 %

Module Quality Loss

Array #1 - Champ PV tiotures
Loss Fraction -0.38 %

Array #2 - Champ BIPV -51
Loss Fraction 2.50 %

Array #3 - Champ BIPV 39
Loss Fraction 2.50 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

DC wiring losses

Global wiring resistance	4.3 mΩ
Loss Fraction	1.50 % at STC

Array #1 - Champ PV tiotures

Global array res.	36 mΩ
Loss Fraction	1.50 % at STC

Array #2 - Champ BIPV -51

Global array res.	8.9 mΩ
Loss Fraction	1.50 % at STC

Array #3 - Champ BIPV 39

Global array res.	11 mΩ
Loss Fraction	1.50 % at STC

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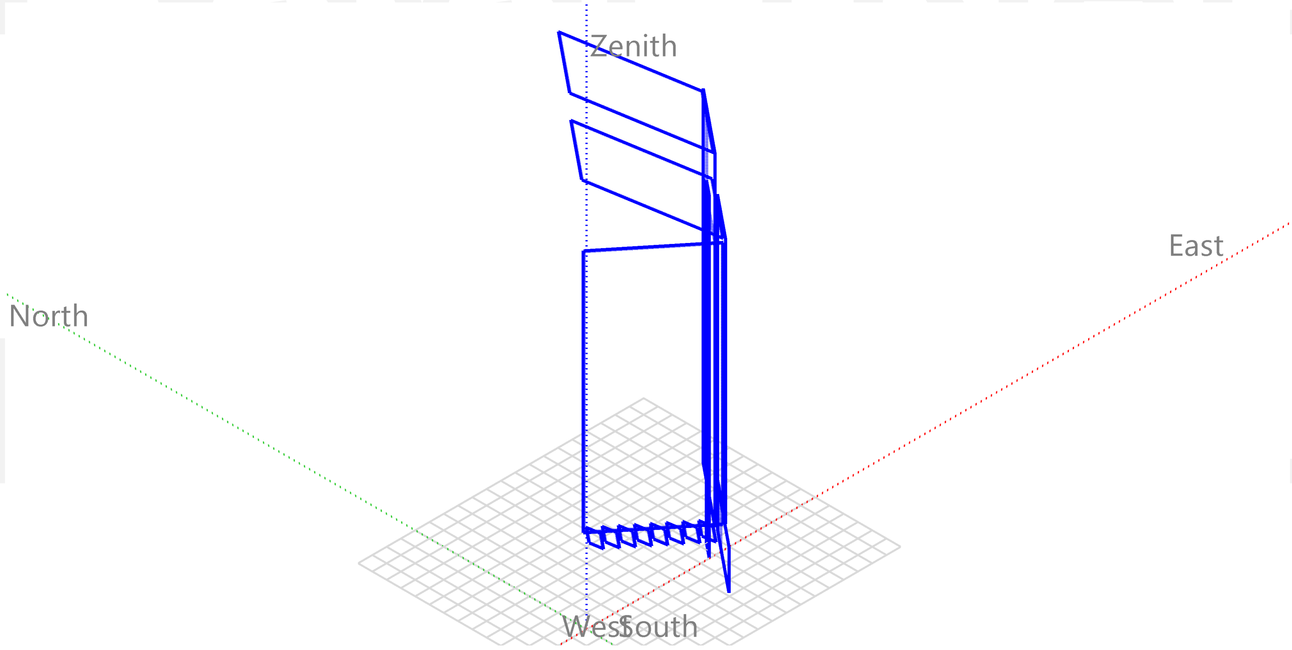
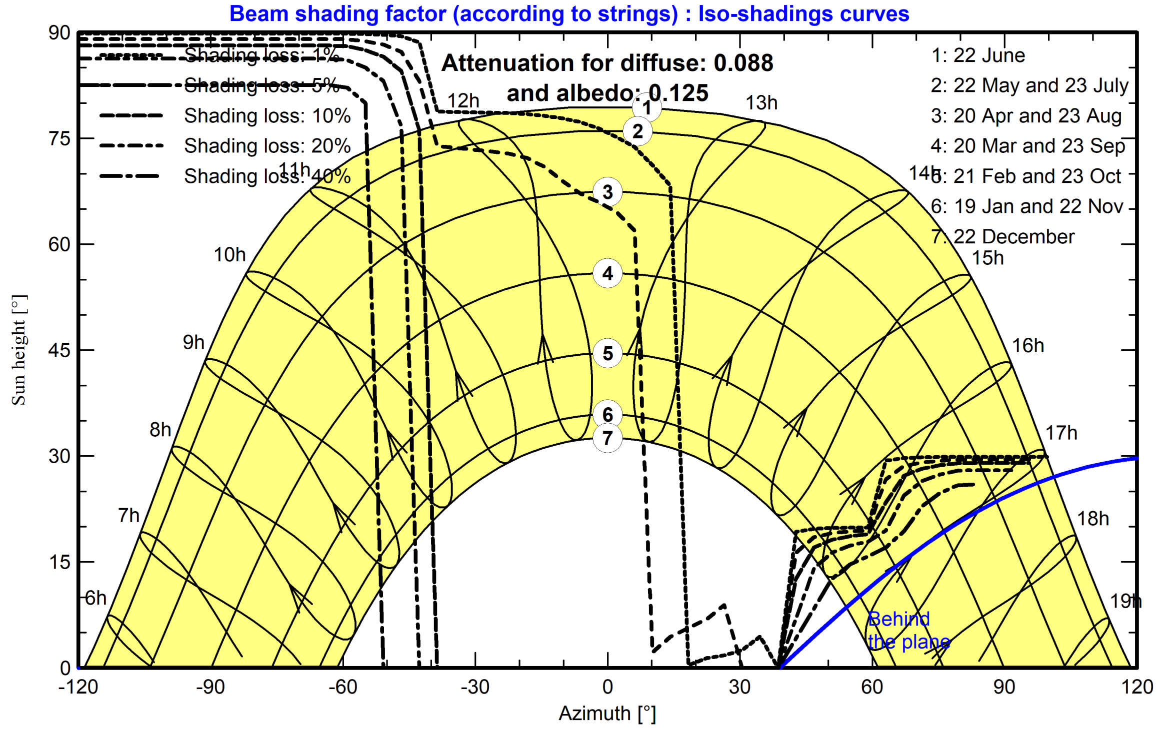
Near shadings parameter

Perspective of the PV-field and surrounding shading scene

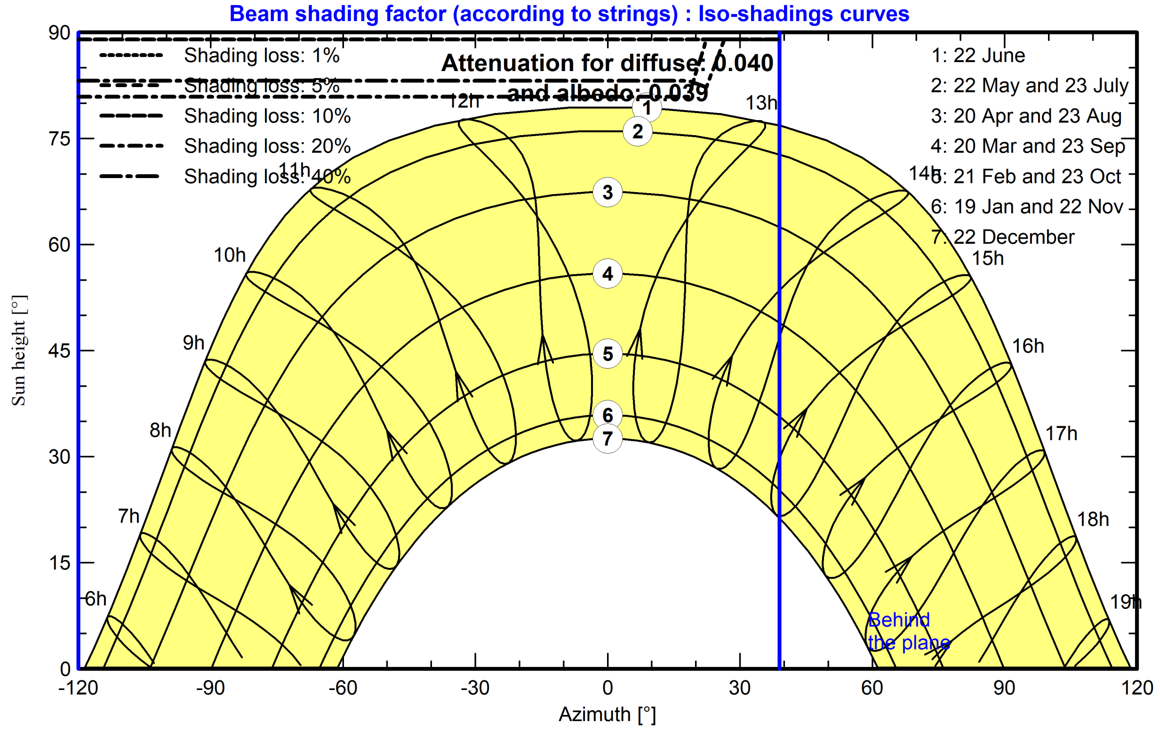


Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 30°/ -51°



Orientation #2 - Fixed plane, Tilts/azimuths: 90°/ -51°



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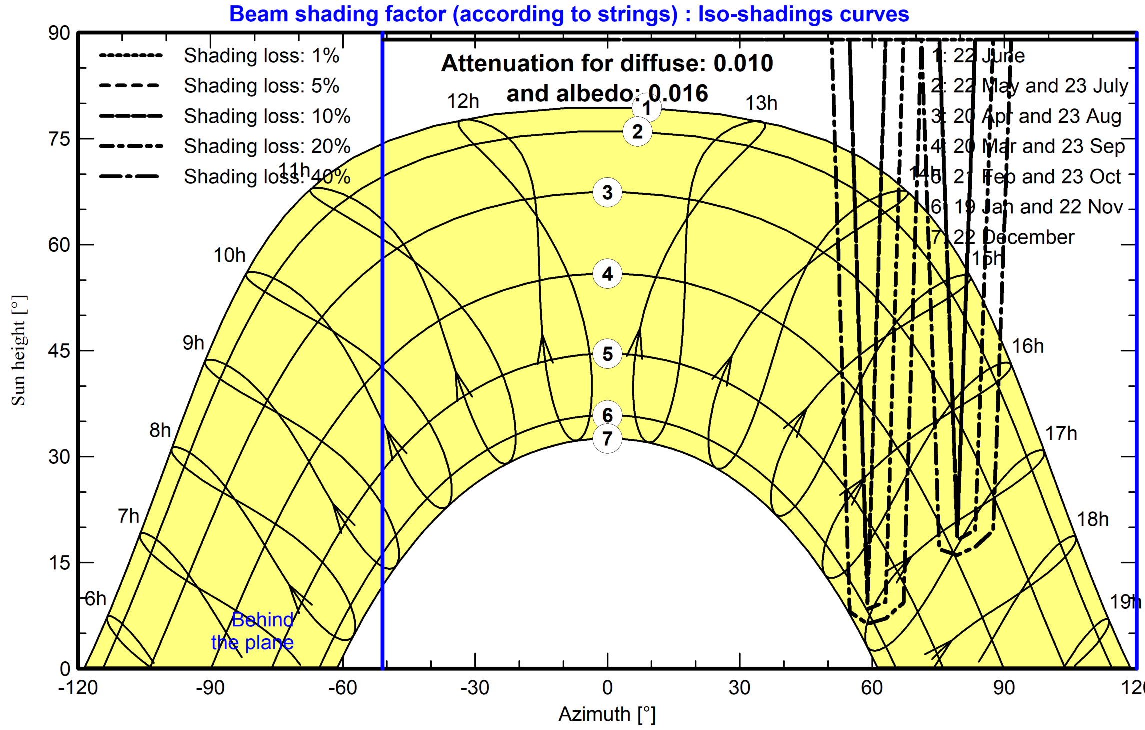
Iso-shadings diagram

Orientation #3 - Fixed plane, Tilts/azimuths: 90°/ 39°



Main results

Simulation step
Hourly

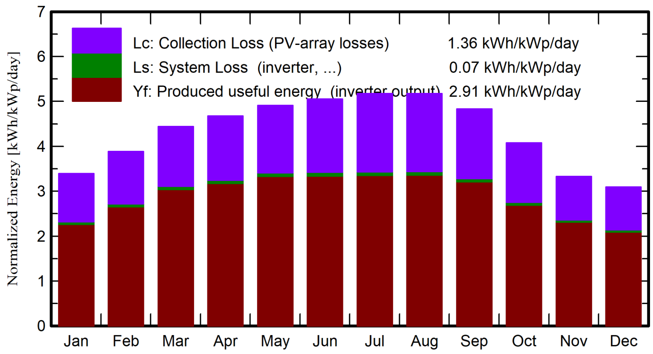
System Production

Produced Energy	910054 kWh/year	Specific production	1061 kWh/kWp/year
		Perf. Ratio PR	67.02 %

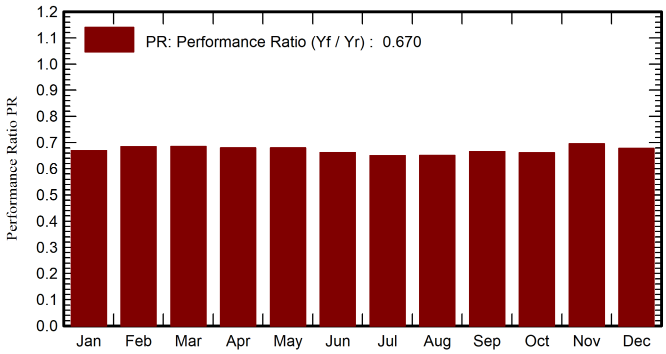
Economic evaluation

Investment		Yearly cost		LCOE	
Global	1 028 889.60 EUR	Annuities	0.00 EUR/yr	Energy cost	0.11 EUR/kWh
Specific	1.20 EUR/Wp	Run. costs	10 416.38 EUR/yr		
		Payback period	16.6 years		

Normalized productions (per installed kWp)



Performance Ratio PR



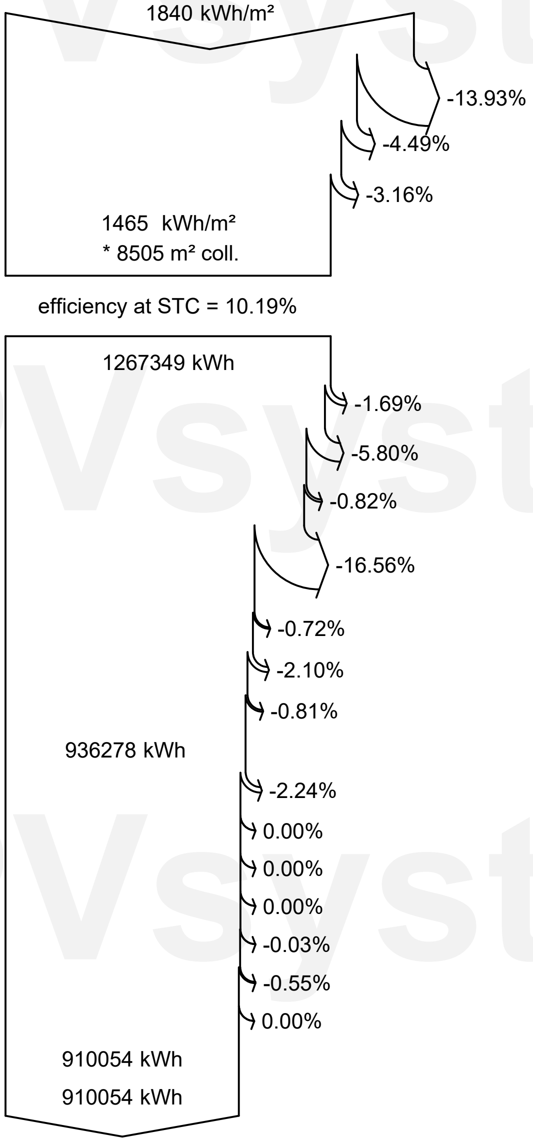
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	86.5	31.72	11.84	105.1	98.4	61784	60301	0.669
February	102.7	39.92	12.64	108.8	102.1	65370	63871	0.684
March	148.8	59.01	14.44	137.7	128.5	82791	80924	0.685
April	177.5	66.56	16.14	140.3	129.7	83577	81692	0.679
May	209.8	81.24	18.64	152.3	139.1	90779	88736	0.679
June	219.6	80.68	21.05	151.6	137.0	88111	86087	0.662
July	228.4	75.43	22.55	160.3	145.7	91380	89292	0.650
August	208.7	69.70	23.15	160.3	147.7	91512	89501	0.651
September	165.7	63.24	21.75	144.9	134.9	84598	82717	0.666
October	127.1	46.13	19.45	126.4	118.5	73312	71625	0.661
November	88.0	42.06	15.44	99.9	93.6	60952	59543	0.695
December	77.2	33.37	13.14	95.8	89.7	57125	55765	0.679
Year	1840.0	689.06	17.55	1583.6	1464.7	931290	910054	0.670

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Spectral correction for amorphous

Shadings: Electrical Loss acc. to strings

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

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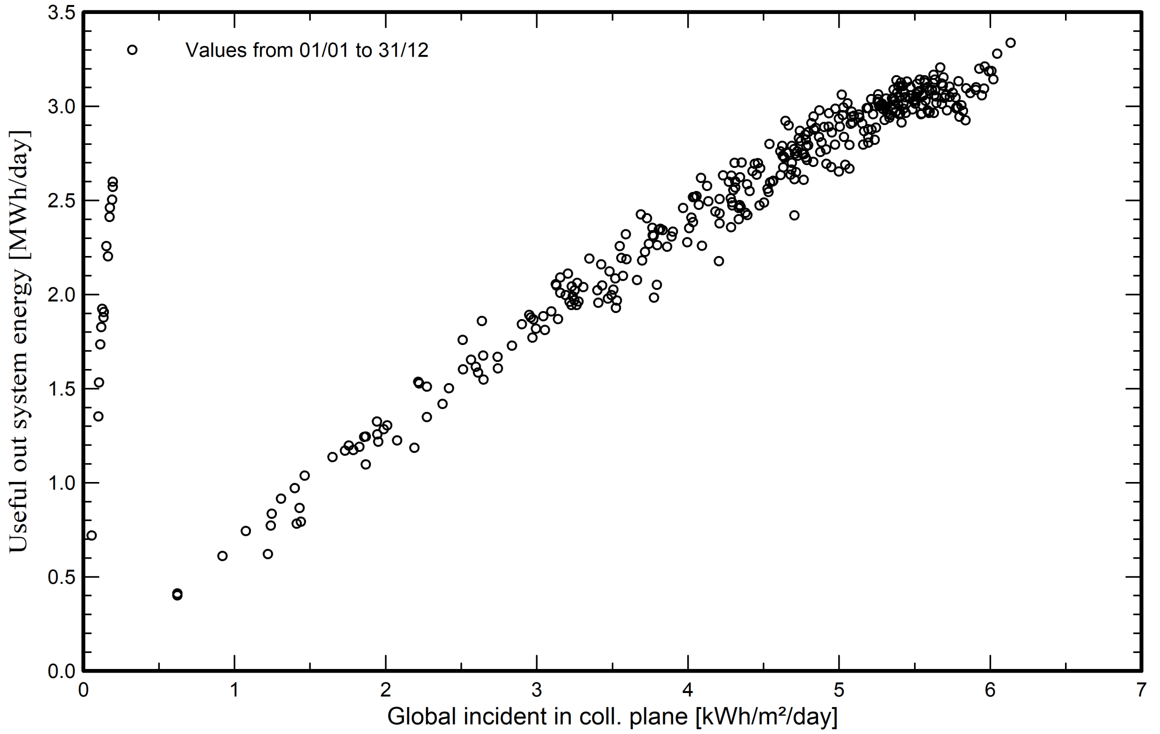
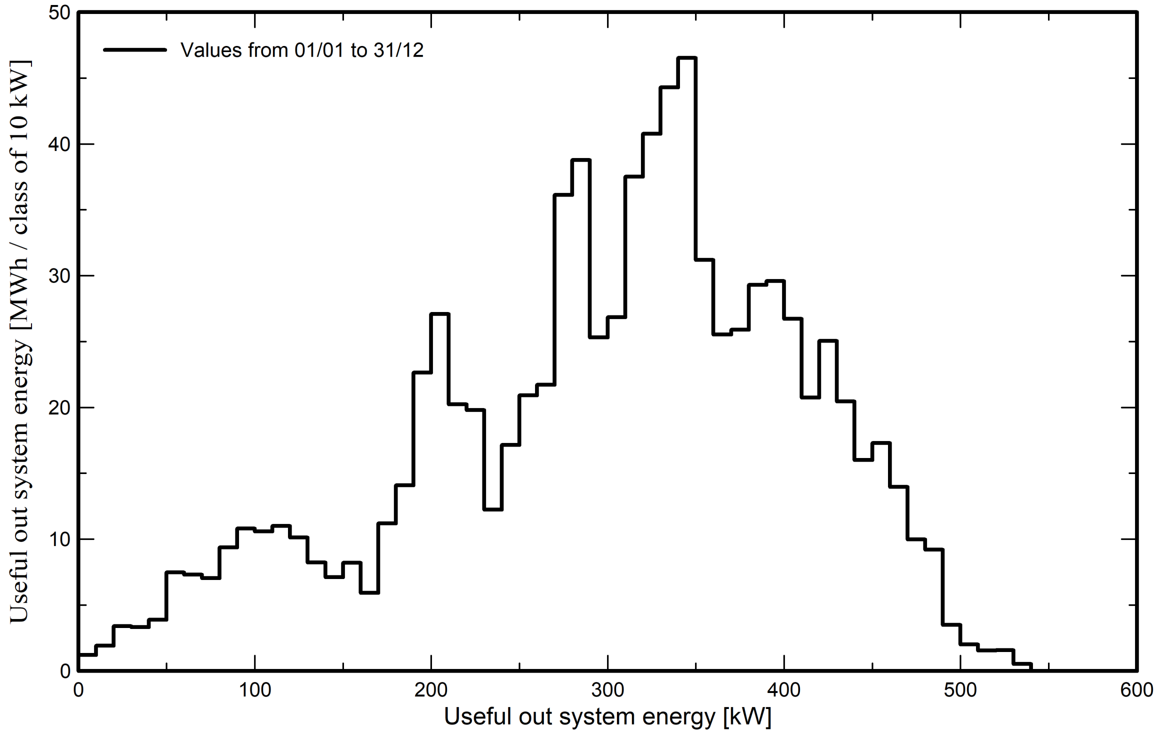
Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
PV modules			
Mono 440 Wp Twin 144 half-cells	936	201.53	188 629.76
a-Si:H, tripple junction	1903	157.69	300 092.80
Micro-crys/Amorphous 115Wc	1624	184.79	300 092.80
Supports for modules	4463	9.61	42 870.40
Inverters			
350 kWac inverter, 12 MPPT	1	31 658.14	34 296.32
12 kWac inverter with 2 MPPT	31	1 936.08	60 018.56
Other components			
Wiring	1	25 722.24	25 722.24
Installation			
Global installation cost per module	4463	13.45	60 018.56
Grid connection	1	17 148.16	17 148.16
		Total	1 028 889.60
		Depreciable asset	926 000.64

Operating costs

Item	Total EUR/year
Maintenance	
Cleaning	8 574.08
Total (OPEX)	8 574.08
Including inflation (2.00%)	10 416.38

System summary

Total installation cost	1 028 889.60 EUR
Operating costs (incl. inflation 2.00%/year)	10 416.38 EUR/year
Produced Energy	910 MWh/year
Cost of produced energy (LCOE)	0.1096 EUR/kWh

Financial analysis

Simulation period

Project lifetime	20 years	Start year	2027

Income variation over time

Inflation	2.00 %/year
Module Degradation	0.00 %/year
Discount rate	6.00 %/year

Income dependent expenses

Income tax rate	20.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (EUR)	Depreciable (EUR)
PV modules				
Mono 440 Wp Twin 144 half-cells	Straight-line	20	0.00	188 629.76
a-Si:H, tripple junction	Straight-line	20	0.00	300 092.80
Micro-crys/Amorphous 115Wc	Straight-line	20	0.00	300 092.80
Supports for modules	Straight-line	20	0.00	42 870.40
Inverters				
350 kWac inverter, 12 MPPT	Straight-line	20	0.00	34 296.32
12 kWac inverter with 2 MPPT	Straight-line	20	0.00	60 018.56
		Total	0.00	926 000.64

Financing

Own funds	1 028 889.60 EUR

Electricity sale

Feed-in tariff	0.11000 EUR/kWh
Duration of tariff warranty	20 years
Annual connection tax	0.00 EUR/year
Annual tariff variation	+3.0 %/year
Feed-in tariff decrease after warranty	0.00 %

Return on investment

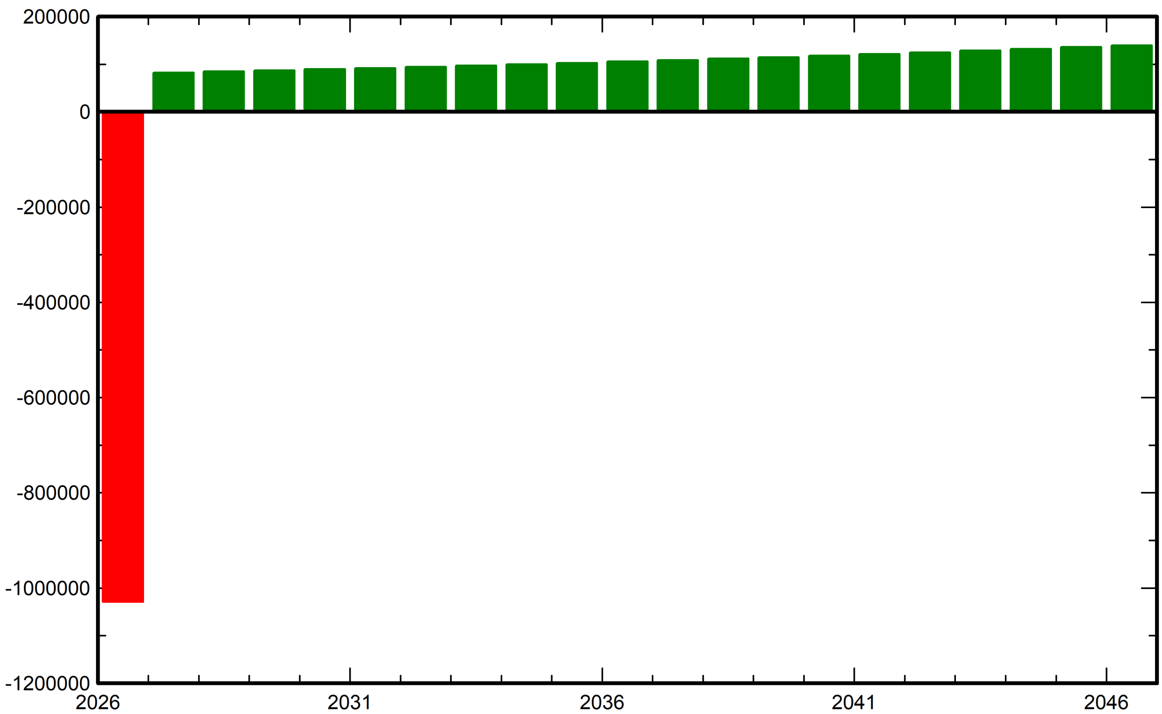
Payback period	16.6 years
Net present value (NPV)	151 460.71 EUR
Internal rate of return (IRR)	7.59 %
Return on investment (ROI)	14.7 %

Financial analysis

Detailed economic results (EUR)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	1 028 890	0	0	0	0	0	-1 028 890	0.0%
1	100 107	0	8 574	46 300	45 233	9 047	82 486	-951 072	7.6%
2	103 110	0	8 746	46 300	48 065	9 613	84 752	-875 644	14.9%
3	106 203	0	8 920	46 300	50 983	10 197	87 086	-802 524	22.0%
4	109 390	0	9 099	46 300	53 991	10 798	89 493	-731 638	28.9%
5	112 671	0	9 281	46 300	57 090	11 418	91 972	-662 911	35.6%
6	116 051	0	9 466	46 300	60 285	12 057	94 528	-596 272	42.0%
7	119 533	0	9 656	46 300	63 577	12 715	97 162	-531 654	48.3%
8	123 119	0	9 849	46 300	66 970	13 394	99 876	-468 991	54.4%
9	126 812	0	10 046	46 300	70 467	14 093	102 673	-408 218	60.3%
10	130 617	0	10 247	46 300	74 070	14 814	105 556	-349 277	66.1%
11	134 535	0	10 452	46 300	77 784	15 557	108 527	-292 106	71.6%
12	138 571	0	10 661	46 300	81 611	16 322	111 589	-236 650	77.0%
13	142 729	0	10 874	46 300	85 555	17 111	114 744	-182 854	82.2%
14	147 010	0	11 091	46 300	89 619	17 924	117 995	-130 664	87.3%
15	151 421	0	11 313	46 300	93 807	18 761	121 346	-80 031	92.2%
16	155 963	0	11 540	46 300	98 124	19 625	124 799	-30 904	97.0%
17	160 642	0	11 770	46 300	102 572	20 514	128 358	16 763	101.6%
18	165 462	0	12 006	46 300	107 156	21 431	132 025	63 017	106.1%
19	170 425	0	12 246	46 300	111 879	22 376	135 804	107 902	110.5%
20	175 538	0	12 491	46 300	116 747	23 349	139 698	151 461	114.7%
Total	**2 689 911**	**1 028 890**	**208 328**	**926 001**	**1 555 583**	**311 117**	**2 170 467**	**151 461**	**114.7%**

Yearly net profit (EUR)



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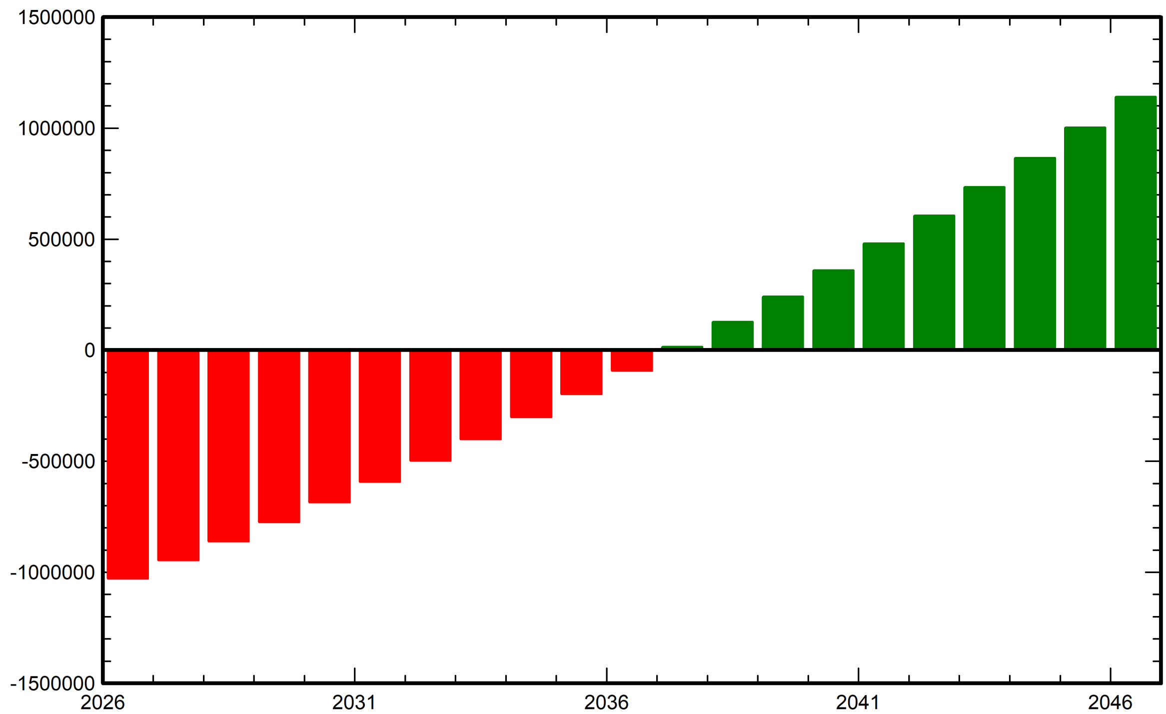
Financial analysis

Cumulative cashflow (EUR)



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CO_2 Emission Balance

Total:	10938.4 tCO_2

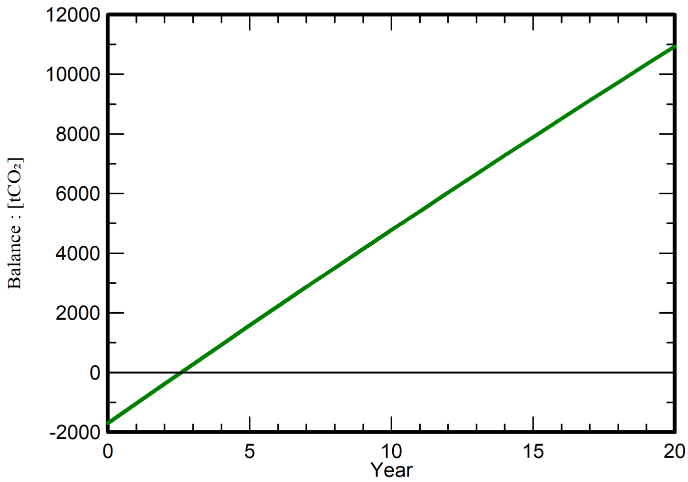
Generated emissions

Total:	1701.09 tCO_2
Source: Detailed calculation from table below	

Replaced Emissions

Total:	13250.4 tCO_2
System production:	910.05 MWh/yr
Grid Lifecycle Emissions:	728 gCO_2/kWh
Source:	IEA List
Country:	Morocco
Lifetime:	20 years
Annual degradation:	0.5 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	857 kWp	1468500
Supports	4.86 kgCO2/kg	44630 kg	216713
Inverters	481 kgCO2/units	33.0 units	15875